SEC P vs NP log 2026-05-21

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-21 08:33 UTC

SEC P vs NP — daily log 2026-05-21

23 cycles recorded

Noncommutative Algebraic Growth and Tseitin Resolution Width

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Noncommutative Algebra × Resolution Width of Tseitin Formulas
- **Recorded**: 2026-05-21 00:30 UTC
- **Entry ID**: `e87da145bc32`

Statement

For a random 3-CNF formula Φ with n variables, let A_Φ be the noncommutative algebra generated by variables x_i and relations x_i x_j = x_j x_i if clauses (i,j) are satisfied. The growth rate of dim(A_Φ_k) is Ω(2^{k/2}) for all k ≤ log n iff Φ's Tseitin formula has resolution width ≥ log n.

Rationale

Noncommutative algebras encode clause interactions via generators, while resolution width measures proof complexity. The conjecture links algebraic growth rates to proof complexity via commutativity constraints, revealing structural trade-offs between algebraic dependencies and proof efficiency.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `27.65s`


```
0.0, 'instances_tested': 10, 'conjecture_holds': False, 'counterexample': 'growth_rate_mismatch'}
TRIAL: {'metric_name': 'growth_rate', 'metric_value': 0.0, 'instances_tested': 10, 'conjecture_holds
TRIAL: {'metric_name': 'growth_rate', 'metric_value': 0.0, 'instances_tested': 10, 'conjecture_holds
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TRIAL: {'metric_name': 'growth_rate', 'metric_value': 0.0, 'instances_tested': 10, 'conjecture_holds
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TRIAL: {'metric_name': 'growth_rate', 'metric_value': 0.0, 'instances_tested': 10, 'conjecture_holds
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TRIAL: {'metric_name': 'growth_rate', 'metric_value': 0.0, 'instances_tested': 10, 'conjecture_holds
TRIAL: {'metric_name': 'growth_rate', 'metric_value': 0.0, 'instances_tested': 10, 'conjecture_holds
```

```
TRIAL: {'metric_name': 'growth_rate', 'metric_value': 0.0, 'instances_tested': 10, 'conjecture_holds
TRIAL: {'metric_name': 'growth_rate', 'metric_value': 0.0, 'instances_tested': 10, 'conjecture_holds
RESULT: FALSIFIED counterexample="growth_rate_mismatch" first_failing_seed=0
```

Judge reasoning

Safety rail: critic_challenge_falsified | original: The conjecture was falsified by a counterexample showing growth_rate_mismatch in all tested instances. | next: Analyze the specific counterexample to determine how the growth rate deviation correlates with Tseitin resolution width properties

---

Noncommutative Rank Gap in Read-Twice Branching Programs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Noncommutative Algebra × Read-Twice Branching Programs
- **Recorded**: 2026-05-21 00:52 UTC
- **Entry ID**: `f245272528b8`

Statement

For a read-twice BP P over n variables, the noncommutative rank of its transition matrix M satisfies rank_nc(M) ≥ Ω(n), whereas for read-once BPs, rank_nc(M) = O(log n).

Rationale

Noncommutative rank captures algebraic dependencies in matrix factorizations, which may reflect the repeated variable access in read-twice BPs. This could expose structural asymmetries in their computational power compared to read-once BPs.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ae8bb1cc.py", line 105, in <module>
    trial_result = run_trial(seed)
                   ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ae8bb1cc.py", line 78, in run_trial
    rank = rank_nc(M, p)
           ^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ae8bb1cc.py", line 38, in rank_nc
    gaussian_elimination(A, p)
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ae8bb1cc.py", line 23, in gaussian_elim
    if abs(A[j][i]) > abs(A[max_row][i]):
       ^^^^^^^^^^^^
TypeError: bad operand type for abs(): 'list'
```

Judge reasoning

Test crashed due to TypeError in Gaussian elimination, preventing result collection. Support condition cannot be evaluated without successful trials. | next: Debug Gaussian elimination implementation to handle matrix entries correctly (ensure numerical values instead of lists)

Non-Abelian Harmonic Analysis and SOS Refutation Degree for CSPs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Non-Abelian Harmonic Analysis × SOS Refutation Degree for CSPs
- **Recorded**: 2026-05-21 01:05 UTC
- **Entry ID**: `93e2fe31dd48`

Statement

For a CSP instance over a non-abelian group G, the SOS refutation degree is bounded by the dual norm of the Fourier coefficients of the CSP's constraint function. Specifically, for every CSP instance φ over G, refutation_degree(φ) ≤ ||F||_∞, where F is the Fourier transform of φ's constraint function.

Rationale

Non-abelian harmonic analysis decomposes CSP constraints into group characters, revealing structural symmetries that may influence the SOS hierarchy's ability to certify unsatisfiability. The dual norm of Fourier coefficients quantifies the 'spread' of constraints across irreducible representations, which could correlate with the minimal SOS degree required for refutation.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6c3843cd.py", line 74, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6c3843cd.py", line 50, in run_trial
    F = [1] * (2 ** n)  # Placeholder Fourier coefficients
        ~~~~^~~~~~~~~~
MemoryError
```

Judge reasoning

Test crashed with MemoryError before producing results, preventing evaluation of support fraction or counterexamples. | next: Optimize memory usage in test code for large n, then re-run experiments

---

Non-Abelian Fourier Coefficient Gap Bounds Disjointness Communication Complexity

- **Verdict**: `INCONCLUSIVE`

- **Bridge**: Non-Abelian Harmonic Analysis × Communication Complexity of Disjointness
- **Recorded**: 2026-05-21 01:53 UTC
- **Entry ID**: `324ca13de3e5`

Statement

For any 3-CNF formula φ with n variables, the minimal non-Abelian Fourier coefficient |F[π]| over all irreducible representations π of the symmetric group S_n satisfies |F[π]| ≥ Ω(1/√n) if and only if the disjointness communication complexity of φ is Ω(log n).

Rationale

Non-Abelian Fourier analysis captures group-theoretic symmetries in communication matrices. Disjointness requires coordination across independent sets, which may manifest as nontrivial Fourier coefficients in symmetric group representations, revealing structural constraints on protocol efficiency.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6de39bfc.py", line 65, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6de39bfc.py", line 45, in run_trial
    clause = [random.randint(0, n-1) for _ in range(3)]
                                ^
NameError: name 'n' is not defined
```

Judge reasoning

Test crashed due to undefined variable 'n' before producing results | next: Fix the NameError in test code by defining 'n' before using it in clause generation

---

ε-Approximation Complexity Lower Bound for k-CLIQUE via Monotone DNF Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Approximation Theory of Boolean Functions × Monotone DNF Formulas for k-CLIQUE
- **Recorded**: 2026-05-21 02:10 UTC
- **Entry ID**: `5c00c6af4f62`

Statement

For any ε ∈ (0, 1/2), the ε-approximation complexity of the k-CLIQUE function is Ω(n) when k ≥ log n, and this complexity is submodular under conjunction of functions.

Rationale

Approximation theory provides tools to analyze how well boolean functions can be approximated by simpler functions, which could reveal structural limitations in monotone DNF representations. The submodularity under conjunction suggests that combining functions increases the required complexity, aligning with known lower bounds for k-CLIQUE.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e72ffab8.py", line 70, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e72ffab8.py", line 49, in run_trial
    clique = generate_k_clique(n, k)
             ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e72ffab8.py", line 25, in generate_k_cl
    if all(node not in edge for edge in clique):
       ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e72ffab8.py", line 25, in <genexpr>
    if all(node not in edge for edge in clique):
           ^^^^^^^^^^^^^^^^
TypeError: argument of type 'int' is not iterable
```

Judge reasoning

Test crashed with TypeError due to non-iterable integer in membership check, preventing data collection | next: Fix the generate_k_clique function to ensure edges are iterable structures (e.g., tuples) before membership testing

---

Matroid Rank and ABP Size for Polynomial Representations

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Matroid Theory × Algebraic Branching Program Size
- **Recorded**: 2026-05-21 02:20 UTC
- **Entry ID**: `d2edbef09287`

Statement

The matroid rank of the coefficient matrix of a polynomial represented by an ABP of size s is Θ(log s).

Rationale

Matroid rank captures the linear independence structure of the polynomial's coefficients, which may reflect the minimal complexity of representing the polynomial via an ABP. This bridge could expose hidden algebraic dependencies in the ABP's structure.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2b11b003.py", line 75, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2b11b003.py", line 51, in run_trial
    A = [[coefficients[i] * j**k for k in range(n + 1)] for i in range(n + 1)]
                            ^
NameError: name 'j' is not defined
```

Judge reasoning

Test crashed due to undefined variable 'j', preventing data collection. Pre-registered support condition cannot be evaluated without successful trials. | next: Fix the NameError in test_2b11b003.py by defining variable 'j' in the list comprehension

---

Sheaf Cohomology Euler Characteristic and Tseitin Resolution Proof Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Sheaf Cohomology × Resolution Proof Size for Tseitin Formulas
- **Recorded**: 2026-05-21 02:31 UTC
- **Entry ID**: `62b0c1d5061d`

Statement

For a Tseitin formula derived from a graph G with n vertices, the absolute value of the Euler characteristic of the sheaf cohomology groups H^i(G) is inversely proportional to the resolution proof size, i.e., |χ(H^i(G))| = Θ(1 / log n).

Rationale

Sheaf cohomology captures the topological structure of the graph, which may influence the complexity of the resolution proof. The Euler characteristic provides a quantitative measure of this topological invariant, potentially linking it to the proof size.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7ff24d2c.py", line 113, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
```

```
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7ff24d2c.py", line 43, in run_trial
    clauses.append([-literals[i], -literals[j]])
                    ^^^^^^^^^^^^
TypeError: bad operand type for unary -: 'str'
```

Judge reasoning

Test crashed due to type error in clause generation, preventing data collection | next: Fix the TypeError by ensuring literals are integers before applying unary negation

---

Fourier Norm Lower Bound for Disjointness Communication Matrices

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Harmonic Analysis on Locally Compact Groups × Randomized Communication Complexity of Disjointness
- **Recorded**: 2026-05-21 03:12 UTC
- **Entry ID**: `af4613884bf0`

Statement

For any n×n communication matrix M of the Disjointness problem, the L^2 norm of its Fourier transform over the additive group Z_n^2 is Ω(n).

Rationale

Harmonic analysis on finite abelian groups captures the spectral structure of communication matrices, with Fourier norms reflecting the 'spread' of information across frequencies. The Disjointness matrix's inherent combinatorial structure may enforce nontrivial Fourier norms, linking algebraic invariants to communication complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3a7c6574.py", line 70, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3a7c6574.py", line 54, in run_trial
    F = fft_2d(M)
        ^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3a7c6574.py", line 43, in fft_2d
    M_fft[i] = fft([M[i][j] for j in range(n)])
               ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3a7c6574.py", line 34, in fft
    even = fft([a[k] for k in range(0, n, 2)])
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3a7c6574.py", line 34, in fft
    even = fft([a[k] for k in range(0, n, 2)])
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3a7c6574.py", line 34, in fft
    even = fft([a[k] for k in range(0, n, 2)])
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  [Previous line repeated 2 more times]
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3a7c6574.py", line 36, in fft
    T = [cmath.exp(-2j * math.pi * k / n) * odd[k] for k in range(n // 2)]
         ^^^^^
NameError: name 'cmath' is not defined. Did you mean: 'math'? Or did you forget to import 'cmath'?
```

Judge reasoning

Test crashed due to missing cmath import, preventing evaluation of Fourier norm bounds. | next: Fix the test by importing cmath and re-running with proper error handling

---

Laplacian Eigenvalue and Resolution Length of Tseitin Formulas

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Spectral Graph Theory × Resolution Length of Tseitin Formulas
- **Recorded**: 2026-05-21 03:30 UTC
- **Entry ID**: `9e9baa385874`

Statement

For any graph G, the resolution length of the Tseitin formula on G is at least 2^{Ω(1/(1 - λ_2))}, where λ_2 is the second smallest eigenvalue of the Laplacian matrix of G. For graphs where 1 - λ_2 ≥ c (constant c > 0), the resolution length is exponential in 1/(1 - λ_2).

Rationale

The second eigenvalue of the Laplacian (λ_2) captures the expansion properties of the graph. A smaller λ_2 indicates a better expander, leading to longer resolution proofs as the expansion increases, requiring more steps to refute the Tseitin formula.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7ed580cd.py", line 85, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7ed580cd.py", line 57, in run_trial
    λ2 = min(laplacian_eigenvalues(G)[1:])
             ^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7ed580cd.py", line 41, in laplacian_eig
    L = gaussian_elimination(L)
        ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7ed580cd.py", line 28, in gaussian_elim
    A[k][j] += factor * A[i][j]
    ~~~~^^^
```

```
IndexError: list index out of range
```

Judge reasoning

Test crashed with IndexError in Gaussian elimination, preventing data collection. Pre-registered support condition cannot be evaluated without successful trials. | next: Fix the Gaussian elimination implementation to handle matrix indices properly and re-run tests with multiple seeds

---

Moment Matrix Spectral Gap and SOS Degree for Max-CUT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Random Matrix Theory × SOS Degree for Max-CUT
- **Recorded**: 2026-05-21 03:38 UTC
- **Entry ID**: `797c606cf901`

Statement

For any n-variable Max-CUT instance, the smallest eigenvalue of its degree-d moment matrix satisfies $\lambda\_min \geq \Theta(1/\sqrt{n})$ iff $d \geq \Omega(\log n)$. The spectral gap $\gamma = \lambda\_max - \lambda\_min$ scales as $\gamma = \Theta(1/n)$ for $d = \Omega(\log n)$.

Rationale

Random matrix theory provides tools to analyze the spectral properties of moment matrices, which are central to SOS hierarchies. The spectral gap directly influences the convergence rate of SOS approximations, linking algebraic structure to algorithmic complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c1d82cae.py", line 180, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c1d82cae.py", line 153, in run_trial
    moment_matrix = construct_moment_matrix(instance, d)
                    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c1d82cae.py", line 39, in construct_mom
    m[i][j] += sum(lit ** abs(j - k) for lit in clause)
               ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c1d82cae.py", line 39, in <genexpr>
    m[i][j] += sum(lit ** abs(j - k) for lit in clause)
                                  ^
NameError: name 'k' is not defined
```

Judge reasoning

Test crashed due to undefined variable 'k' in the moment matrix construction, preventing data collection. | next: Debug the NameError in the genexpr by checking variable scoping for 'k' in the clause summation

---

Free Entropy Gap in Read-Twice BP Transition Matrices

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Free Probability × Read-Twice Branching Programs
- **Recorded**: 2026-05-21 03:52 UTC
- **Entry ID**: `b6ef2369c2bb`

Statement

For a read-twice BP P over n-bit inputs, let μ_P be the non-commutative distribution of its transition matrices. Define ρ(P) = |R(μ_P)(0)|, where R(μ_P) is the R-transform of μ_P. Then ρ(P) ≥ Ω(log n) for all P computing IP_2, but ρ(P) = O(1) for all read-once BPs computing IP_2.

Rationale

Free entropy measures the 'non-commutative complexity' of random variables, which may capture the inherent structure of read-twice BPs that read-once BPs lack. The R-transform's zeroth coefficient encodes global scaling behavior, potentially distinguishing these models via their algebraic invariants.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `8.28s`

```
e': Fraction(1, 1), 'instances_tested': 30, 'conjecture_holds': False, 'counterexample': 'ρ(P) < 3.5'
TRIAL: {'metric_name': 'ρ(P)', 'metric_value': Fraction(1, 1), 'instances_tested': 30, 'conjecture_ho
TRIAL: {'metric_name': 'ρ(P)', 'metric_value': Fraction(1, 1), 'instances_tested': 30, 'conjecture_ho
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TRIAL: {'metric_name': 'ρ(P)', 'metric_value': Fraction(1, 1), 'instances_tested': 30, 'conjecture_ho
TRIAL: {'metric_name': 'ρ(P)', 'metric_value': Fraction(1, 1), 'instances_tested': 30, 'conjecture_ho

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a9497589.py", line 76, in <module>
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
                        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a9497589.py", line 76, in <genexpr>
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
                                  ~^^^^^^^^
KeyError: 'seed'
```

Judge reasoning

Test crashed before producing reliable data, but stdout suggests counterexample ρ(P) < 3.5. The crash invalidates the counterexample's validity. | next: Re-run test with increased instance count and seed diversity to verify conjecture status

---

Matroid Rank Gap in Monotone DNF for k-CLIQUE

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Matroid Theory × Monotone DNF Formulas for k-CLIQUE
- **Recorded**: 2026-05-21 04:29 UTC
- **Entry ID**: `202b457c781c`

Statement

For a k-CLIQUE indicator function expressed as a monotone DNF with m clauses, the rank of the matroid formed by its clause vectors is Ω(n) if m ≤ n^{O(1)}, and O(log n) if m ≥ $n^{\{\Omega(k}$\{1/4\})\}.

Rationale

Matroid rank captures combinatorial dependencies in clause interactions, which may expose structural barriers to succinct DNF representations. The conjecture ties this invariant to known lower bounds for k-CLIQUE, suggesting rank gaps could amplify hardness.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_115ac035.py", line 83, in <module>
    trial_result = run_trial(seed)
                   ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_115ac035.py", line 56, in run_trial
    matrix = [[int(abs(lit) == var) for var in range(1, n + 1)] for clause in dnf]
                       ^^^
NameError: name 'lit' is not defined. Did you mean: 'List'?
```

Judge reasoning

Test crashed due to NameError, preventing data collection. Pre-registered support condition cannot be evaluated without successful trials. | next: Fix the NameError in test_115ac035.py by defining 'lit' variable before usage

---

Algebraic Shifting Facet Count Bounds Tseitin Resolution Length

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic shifting × Resolution length of Tseitin formulas
- **Recorded**: 2026-05-21 04:56 UTC
- **Entry ID**: `312ea3700c09`

Statement

For a Tseitin formula on a graph G, the resolution length is ≥ 2^{Ω(ν(G))}, where ν(G) is the number of facets in the algebraic shifted complex of G. This holds for all graphs with expansion parameter λ(G) ≥ 1/√n.

Rationale

Algebraic shifting captures combinatorial structure of graphs via simplicial complexes, while resolution length measures proof complexity. The conjecture links structural properties of shifted complexes to proof complexity, leveraging the interplay between algebraic topology and proof complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_91e2d45b.py", line 105, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_91e2d45b.py", line 80, in run_trial
    expansion_param = sum(len(list(graph.neighbors(v))) for v in range(n)) / (n * (n - 1))
                      ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_91e2d45b.py", line 80, in <genexpr>
    expansion_param = sum(len(list(graph.neighbors(v))) for v in range(n)) / (n * (n - 1))
                                   ^^^^^^^^^^^^^^^
AttributeError: 'set' object has no attribute 'neighbors'
```

Judge reasoning

Test crashed due to AttributeError in graph.neighbors() call on set objects, preventing data collection | next: Fix graph representation to use a structure with neighbors() method (e.g., NetworkX Graph) and re-run experiments

---

Symmetric Group Orbit Count Distinguishes Read-Twice BPs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Invariant Theory × Read-Twice Branching Programs
- **Recorded**: 2026-05-21 05:31 UTC
- **Entry ID**: `376a7658c1fe`

Statement

For a read-twice BP P, the number of orbits of its associated polynomial representation under the symmetric group S_n action is Θ(log size(P)) for read-once BPs but Ω(n) for read-twice BPs. The orbit count is strictly greater for read-twice BPs when size(P) ≥ 2^n/2.

Rationale

Invariant theory provides a framework to measure symmetry in polynomial representations, which could expose structural differences between read-once and read-twice BPs. The symmetric group action on variables creates a natural invariant that may scale differently with BP complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `245.34s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results; insufficient data to evaluate support fraction or counterexamples. | next: Run test with extended timeout and higher resource allocation

---

Polynomial Threshold Function Degree Bounds Karchmer-Wigderson Protocol Cost for PARITY

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Polynomial Threshold Functions × Karchmer-Wigderson Protocol Cost
- **Recorded**: 2026-05-21 05:41 UTC
- **Entry ID**: `8a180d66cac2`

Statement

For any CNF formula F on n variables computing PARITY, the minimal degree of a polynomial threshold function representing F is Ω(log n).

Rationale

Polynomial threshold functions capture the structure of Boolean functions, and their degree may reflect the complexity of communication protocols needed to compute them. The Karchmer-Wigderson protocol cost measures the communication complexity of computing a function, linking algebraic representation to interactive proof strategies.

Novelty

- Judge: `ADJACENT_OK` over 7 arXiv hits

Top hits consulted: - [2002.07444v1] Multiparty Karchmer-Wigderson Games and Threshold Circuits - [2107.05128v2] Karchmer-Wigderson Games for Hazard-free Computation - [1504.01064v2] The degree of the Alexander polynomial is an upper bound for the topological slice genus - [1806.07358v1] No Threshold graphs are cospectral - [1007.1875v2] Lower Bounds for Quantum Oblivious Transfer

Empirical Test

- exit code: `0`, elapsed: `0.06s`

```
: 1, "conjecture_holds": True, "counterexample": ""}
TRIAL: {"seed": 503, "metric_name": "polynomial_threshold_function_degree", "metric_value": 7.700825
TRIAL: {"seed": 547, "metric_name": "polynomial_threshold_function_degree", "metric_value": 3.529875
TRIAL: {"seed": 593, "metric_name": "polynomial_threshold_function_degree", "metric_value": 2.341542
TRIAL: {"seed": 631, "metric_name": "polynomial_threshold_function_degree", "metric_value": 2.728451
TRIAL: {"seed": 677, "metric_name": "polynomial_threshold_function_degree", "metric_value": 6.381222
TRIAL: {"seed": 727, "metric_name": "polynomial_threshold_function_degree", "metric_value": 7.854839
TRIAL: {"seed": 773, "metric_name": "polynomial_threshold_function_degree", "metric_value": 3.234422
TRIAL: {"seed": 821, "metric_name": "polynomial_threshold_function_degree", "metric_value": 8.674023
TRIAL: {"seed": 877, "metric_name": "polynomial_threshold_function_degree", "metric_value": 1.723483
TRIAL: {"seed": 929, "metric_name": "polynomial_threshold_function_degree", "metric_value": 6.568578
RESULT: SUPPORTED mean=4.618178297706176 std=2.272930228914458 support_fraction=0.9
```

Judge reasoning

parse_fail | next: NONE

Fourier Coefficient Sum Bounds Monotone DNF Size for k-CLIQUE

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Fourier Analysis on Boolean Functions × Monotone DNF Size for k-CLIQUE
- **Recorded**: 2026-05-21 06:02 UTC
- **Entry ID**: `fd209c52f373`

Statement

For any monotone DNF formula on n variables, the sum of absolute values of its Fourier coefficients is O(log n). For the k-CLIQUE indicator function, this sum is Ω(n) for k ≥ 3.

Rationale

Fourier coefficients capture the influence of variables in DNF formulas. Monotone k-CLIQUE requires exponentially many terms, leading to large coefficient sums, while polynomial-size DNFs have logarithmic sums due to sparsity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.04s`

```
', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'Fourier coefficient sum', 'metric_value': 1, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'Fourier coefficient sum', 'metric_value': 1, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'Fourier coefficient sum', 'metric_value': 1, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'Fourier coefficient sum', 'metric_value': 1, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'Fourier coefficient sum', 'metric_value': 1, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'Fourier coefficient sum', 'metric_value': 1, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'Fourier coefficient sum', 'metric_value': 1, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'Fourier coefficient sum', 'metric_value': 1, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'Fourier coefficient sum', 'metric_value': 1, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'Fourier coefficient sum', 'metric_value': 1, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'Fourier coefficient sum', 'metric_value': 1, 'instances_tested': 1, 'conjectu
```

```
TRIAL: {'metric_name': 'Fourier coefficient sum', 'metric_value': 1, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'Fourier coefficient sum', 'metric_value': 1, 'instances_tested': 1, 'conjectu
RESULT: SUPPORTED mean=0.9333333333333333 std=0.24944382578492943 support_fraction=1.0
```

Judge reasoning

All trials used n=1, making the Fourier coefficient sum trivially 1. This violates scaling requirements for both O(log n) and Ω(n) claims. | next: Test with n ≥ 1000 and varying k to observe scaling behavior of Fourier coefficient sums

---

Real Radical Dimension Bounds AC^0 Circuit Size for PARITY

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Real Algebraic Geometry × AC^0 Circuit Size for PARITY
- **Recorded**: 2026-05-21 06:31 UTC
- **Entry ID**: `2c022a5d9d33`

Statement

For any AC^0 circuit C computing PARITY on n inputs, the dimension of the real radical of the ideal generated by C's truth-table polynomial equations satisfies $\dim(\mathrm{rad}(I\_C)) \geq \log_2(\mathrm{size}(C)) - O(\sqrt{n})$.

Rationale

Real radicals capture the algebraic structure of boolean functions; their dimension may encode circuit complexity via elimination theory. PARITY's inherent symmetry could create nontrivial constraints on radical varieties, linking algebraic geometry to circuit size.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `241.79s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results; support fraction cannot be evaluated | next: Run test with extended timeout and verify implementation correctness

---

Eigenvalue Count and SOS Degree for Max-CUT Approximation

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Random Matrix Theory × SOS Degree for Max-CUT
- **Recorded**: 2026-05-21 06:56 UTC
- **Entry ID**: `c0336452dd8d`

Statement

For a random graph $G(n, 1/2)$, the minimal SOS degree required to approximate max-CUT within a factor of $0.878 + \varepsilon$ is $\Omega(\sqrt{n})$, where the constant depends on the number of eigenvalues of G's adjacency matrix in the interval $[-\sqrt{(n \log n)}, \sqrt{(n \log n)}]$.

Rationale

The eigenvalue distribution of random graphs influences the structure of the feasible region for SOS relaxations. Concentration of measure in random matrices suggests that eigenvalues cluster around certain values, which could necessitate higher SOS degrees to capture the necessary precision for approximation.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 5, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 4, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 3, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 4, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 5, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 3, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 5, 'instances_tested': 1, 'conjecture_holds': Tr

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_71ec784e.py", line 71, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_71ec784e.py", line 43, in run_trial
    eigenvalues = gaussian_elimination(adjacency_matrix)
                  ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_71ec784e.py", line 26, in gaussian_elim
    factor_i = 1 / augmented_matrix[i][i]
               ~~^~~~~~~~~~~~~~~~~~~~~~~~
ZeroDivisionError: float division by zero
```

Judge reasoning

Test crashed with ZeroDivisionError, preventing reliable evaluation of the conjecture's validity across sufficient instances. | next: Fix the gaussian_elimination function to handle singular matrices, then re-run tests with more seeds to validate the conjecture

---

Prime Density in Arithmetic Progressions and Seed Length of Nisan-Wigderson PRGs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic Number Theory × Seed Length of Nisan-Wigderson PRGs
- **Recorded**: 2026-05-21 07:37 UTC
- **Entry ID**: `ffe6d4da52cf`

Statement

For a 3-SAT instance with n variables, the seed length of a Nisan-Wigderson PRG that fools AC^0 circuits is $\Theta(\pi(n))$, where $\pi(n)$ is the number of primes $\leq n$. This holds for all instances where the clause graph is bipartite and the variable assignments induce a Dirichlet progression with modulus $\leq n^2$.

Rationale

The distribution of primes in arithmetic progressions relates to the structure of number fields, which could influence the pseudorandomness properties needed for PRGs. The density of primes affects the entropy required to generate pseudorandom bits, thus influencing seed length.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.06s`

```
': 'Seed Length', 'metric_value': None, 'instances_tested': 0, 'conjecture_holds': False, 'counterexa
TRIAL: {'metric_name': 'Seed Length', 'metric_value': None, 'instances_tested': 0, 'conjecture_holds
TRIAL: {'metric_name': 'Seed Length', 'metric_value': None, 'instances_tested': 0, 'conjecture_holds
TRIAL: {'metric_name': 'Seed Length', 'metric_value': None, 'instances_tested': 0, 'conjecture_holds
TRIAL: {'metric_name': 'Seed Length', 'metric_value': None, 'instances_tested': 0, 'conjecture_holds
TRIAL: {'metric_name': 'Seed Length', 'metric_value': None, 'instances_tested': 0, 'conjecture_holds
TRIAL: {'metric_name': 'Seed Length', 'metric_value': None, 'instances_tested': 0, 'conjecture_holds
TRIAL: {'metric_name': 'Seed Length', 'metric_value': None, 'instances_tested': 0, 'conjecture_holds
TRIAL: {'metric_name': 'Seed Length', 'metric_value': None, 'instances_tested': 0, 'conjecture_holds
TRIAL: {'metric_name': 'Seed Length', 'metric_value': None, 'instances_tested': 0, 'conjecture_holds
TRIAL: {'metric_name': 'Seed Length', 'metric_value': None, 'instances_tested': 0, 'conjecture_holds
TRIAL: {'metric_name': 'Seed Length', 'metric_value': None, 'instances_tested': 0, 'conjecture_holds
RESULT: FALSIFIED counterexample="Modulus must be a prime number" first_failing_seed=11
```

Judge reasoning

No instances were tested, violating the 'n too small' failure mode. The counterexample references modulus requirements but lacks empirical validation. | next: Run tests with bipartite clause graphs and prime moduli to collect empirical data on seed length behavior

---

Free Entropy Gap in Disjointness Communication Matrices

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Free Probability × Communication Complexity of Disjointness
- **Recorded**: 2026-05-21 07:51 UTC
- **Entry ID**: `ba26a4ca7784`

Statement

For a random Disjointness matrix $M \in \{0,1\}^{n\times n}$, let $\chi(M)$ denote the free entropy of its entries treated as non-commutative random variables. Then $\chi(M) \geq \Omega(\sqrt{n})$ holds with probability ≥ 0.95 over random seeds, and $\chi(M) = \Theta(\sqrt{n})$ implies the randomized communication complexity of M is $\Omega(n)$.

Rationale

Free entropy quantifies the 'non-commutative complexity' of matrix entries, capturing dependencies that correlate with communication bottlenecks. By framing Disjointness matrices as non-commutative distributions, we expose structural constraints that may lower-bound communication requirements via free probability's asymptotic freeness properties.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
lue': -0.5625, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': 'chi(M) < 4√n'}
TRIAL: {'metric_name': 'free_entropy', 'metric_value': -0.625, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'free_entropy', 'metric_value': -0.625, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'free_entropy', 'metric_value': -0.4375, 'instances_tested': 1, 'conjecture_h
TRIAL: {'metric_name': 'free_entropy', 'metric_value': -0.4375, 'instances_tested': 1, 'conjecture_h
TRIAL: {'metric_name': 'free_entropy', 'metric_value': -0.5, 'instances_tested': 1, 'conjecture_hold
TRIAL: {'metric_name': 'free_entropy', 'metric_value': -0.5625, 'instances_tested': 1, 'conjecture_h
TRIAL: {'metric_name': 'free_entropy', 'metric_value': -0.5625, 'instances_tested': 1, 'conjecture_h
TRIAL: {'metric_name': 'free_entropy', 'metric_value': -0.375, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'free_entropy', 'metric_value': -0.5, 'instances_tested': 1, 'conjecture_hold

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a045ef7b.py", line 60, in <module>
    first_failing_seed = next(res["seed"] for res in results if not res["conjecture_holds"])
                         ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a045ef7b.py", line 60, in <genexpr>
    first_failing_seed = next(res["seed"] for res in results if not res["conjecture_holds"])
                                  ~~~^^^^^^^^
KeyError: 'seed'
```

Judge reasoning

Test crashed before completing validation; counterexamples show failure but cannot trust results due to crash | next: Fix seed handling in test code and re-run with sufficient sample size

---

SOS Moment Matrix Spectral Gap and Max-CUT Approximation Ratio

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Real Algebraic Geometry × SOS Degree for Max-CUT
- **Recorded**: 2026-05-21 08:09 UTC
- **Entry ID**: `49361c02c035`

Statement

For every Max-CUT instance with n variables, if the degree-d SOS moment matrix M_d has minimal eigenvalue λ_min(M_d) ≥ ε, then the integrality gap of the d-round SOS relaxation is ≤ 1 - ε. Conversely, if the integrality gap exceeds 0.878, then λ_min(M_d) < 1 - 0.878 for all d ≤ log n.

Rationale

The spectral gap of the moment matrix encodes the SOS relaxation's ability to capture low-degree certificates. By linking eigenvalue thresholds to approximation ratios, we expose structural constraints between algebraic geometry and optimization hardness, avoiding algebrization barriers through explicit spectral analysis.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.87s`

```
value': 1.0142857142857142, 'instances_tested': 3, 'conjecture_holds': False, 'counterexample': 'Gap
epsilon=5/2'}
TRIAL: {'metric_name': 'Integrality Gap', 'metric_value': 1.0714285714285714, 'instances_tested': 3,
epsilon=5/2'}
TRIAL: {'metric_name': 'Integrality Gap', 'metric_value': 1.0059751773049646, 'instances_tested': 3,
epsilon=525/188'}
TRIAL: {'metric_name': 'Integrality Gap', 'metric_value': 1.0130870210295446, 'instances_tested': 3,
epsilon=23245/5392'}
TRIAL: {'metric_name': 'Integrality Gap', 'metric_value': 1.0416666666666667, 'instances_tested': 3,
epsilon=5/2'}
TRIAL: {'metric_name': 'Integrality Gap', 'metric_value': 1.004379479878373, 'instances_tested': 3,
epsilon=131545706/32251283'}
TRIAL: {'metric_name': 'Integrality Gap', 'metric_value': 1.01875, 'instances_tested': 3, 'conjecture
epsilon=13/4'}
TRIAL: {'metric_name': 'Integrality Gap', 'metric_value': 1.0042481778622288, 'instances_tested': 3,
epsilon=16057508/4367783'}
TRIAL: {'metric_name': 'Integrality Gap', 'metric_value': 1.0113157894736842, 'instances_tested': 3,
epsilon=587/200'}
TRIAL: {'metric_name': 'Integrality Gap', 'metric_value': 1.0110294117647058, 'instances_tested': 3,
epsilon=5/2'}
RESULT: FALSIFIED counterexample="Gap exceeds 1 - epsilon" first_faili
```

Judge reasoning

parse_fail | next: NONE

Additive Energy of Fourier Coefficients and ACC⁰ Circuit Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Additive Combinatorics × ACC⁰ Circuit Size
- **Recorded**: 2026-05-21 08:19 UTC
- **Entry ID**: `2088c7f83a48`

Statement

For every Boolean function f: {0,1}^n → {0,1} in P, the additive energy E(f) of its Fourier coefficients satisfies E(f) ≥ 2^{Ω(n)}, whereas for ACC⁰ circuits computing f, E(f) ≤ $2^{\{O(\log}$2 n)}.

Rationale

Additive energy captures the structure of Fourier coefficients, which reflect the correlation of f with low-degree polynomials. ACC^0 circuits, constrained by bounded-depth, exhibit limited structure, leading to lower additive energy. This could expose a gap between P and ACC^0 via Fourier-analytic methods.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c1fe85aa.py", line 69, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c1fe85aa.py", line 47, in run_trial
    transformed_f = fast_walsh_hadamard_transform(f)
                    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c1fe85aa.py", line 26, in fast_walsh_ha
    v = f[j + m]
        ~^^^^^^^
IndexError: list index out of range
```

Judge reasoning

Test crashed during execution, preventing evaluation of the conjecture's validity. | next: Debug the fast_walsh_hadamard_transform implementation to fix index out of range error

---

Betti Number Sum Lower Bound for SOS Max-CUT Degree

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic Topology × SOS Degree for Max-CUT
- **Recorded**: 2026-05-21 08:33 UTC
- **Entry ID**: `b782e0cb2c69`

Statement

For a graph G on n vertices, the SOS degree required to approximate Max-CUT with 0.878-approximation ratio is at least the sum of the Betti numbers $\beta_0 + \beta_1$ of G's clique complex. This bound is tight for complete graphs.

Rationale

Betti numbers encode topological complexity of the clique complex, which may correlate with the algebraic structure of the Max-CUT instance. The SOS hierarchy's degree reflects the need to resolve higher-dimensional interactions, potentially mirroring the topological complexity captured by Betti numbers.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.03s`

```
e': ''}
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 2, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 2, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': Tr
RESULT: SUPPORTED mean=1.0666666666666667 std=0.24944382578492943 support_fraction=1.0
```

Judge reasoning

All trials used n=1 graphs where Betti numbers are trivial, making the conjecture vacuous. SOS degree is capped at 1, preventing meaningful validation of the lower bound. | next: Test conjecture on graphs with n ≥ 4 to observe non-trivial Betti number interactions